GroupI_HM1

Flego, Gigante, Norbedo, Ortoleva, Mutua

2026-10-31

Contents

FSDS - Chapter 2 . . . 1
Ex 2.8 . . . 1
Ex 2.16 . . . 2
Ex 2.21 . . . 4
Ex 2.26 . . . 4
Ex 2.52 . . . 5
Ex. 2.53 . . . 6
Ex 2.70 . . . 7
FSDS - Chapter 3 . . . 11
Ex 3.18 . . . 11
Ex 3.28 . . . 12
Ex 3.24 . . . 12
FSDS - Chapter 4 . . . 12
Ex 4.14 . . . 12
Ex 4.16 . . . 14
Ex 4.48 . . . 15
FSDS - Chapter 5 . . . 15
Ex 5.2 . . . 15
Ex 5.12 . . . 16
Ex 5.50 . . . 16
CS - Chapter 1 . . . 17
Ex 1.1 . . . 17
Ex 1.6 . . . 18
CS - Chapter 3 . . . 18
Ex 3.5 . . . 18
Ex 3.6 . . . 19

FSDS - Chapter 2

Ex 2.8

Each time a person shops at a grocery store, the event of catching a cold or some other virus from another shopper is independent from visit to visit and has a constant probability over the year, equal to 0.01.

(a) In 100 trips to this store over the course of a year, the probability of catching a virus while shopping there is $100(0.01) = 1.0$. What is wrong with this reasoning?
(b) Find the correct probability in (a).

Solution

(a) The formula at point (a) is the formula for the expected value of a Binomial distribution ($\mathbb{E}(X) = np$). Summing the event probabilities is allowed only if the events are mutually exclusive, but in this case they are

independent. Furthermore, if we visited the store 150 times, the probability would be 1.5, which is impossible.

(b) The actual probability, for 100 independent events with constant probability, is

$$P(X \geq 1) = P(X = 1) + \cdots + P(X = 100)$$

Since it is too long to compute directly, we can simply compute

$$P(X \geq 1) = 1 - P(X = 0)$$

that is to say, 1 minus the probability of never catching a virus. The probability of catching the virus in one visit is $p = 0.01$, so the probability of not catching it in one visit is $1 - p = 0.99$. Since we are dealing with independent events, the probability of not catching the virus in all 100 visits is simply:

$$P(X = 0) = (1 - p)^{100} = (0.99)^{100}$$

```
n <- 100
p <- 0.01
p_0 <- (1 - p)**n
p_ge1 <- 1 - p_0
p_ge1
```

```
## [1] 0.6339677
```

Finally, the probability of catching the virus at least once in 100 visits is:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.99)^{100} \approx 0.634$$

Ex 2.16

Each day a hospital records the number of people who come to the emergency room for treatment.

(a) In the first week, the observations from Sunday to Saturday are 10, 8, 14, 7, 21, 44, 60. Do you think that the Poisson distribution might describe the random variability of this phenomenon adequately? Why or why not?

(b) Would you expect the Poisson distribution to better describe, or more poorly describe, the number of weekly admissions to the hospital for a rare disease? Why?

Solution

(a) No. A Poisson distribution has a single constant rate λ, with

$$E[X] = \text{Var}(X) = \lambda$$

so the variance-to-mean ratio should be close to 1.

```
x <- c(10, 8, 14, 7, 21, 44, 60)
mean(x)
```

```
## [1] 23.42857
```

```
var(x)
```

```
## [1] 423.9524
```

```
var(x) / mean(x)
```

```
## [1] 18.09553
```

The variance is about 18 times the mean so the rate is not constant.

(b) Better. For a rare disease in a very large population, each person has a small probability p of being admitted, so the weekly count is $\text{Bin}(n, p)$. For n large and p small, this is approximately $\text{Pois}(np)$, provided the cases are independent and the rate is roughly constant. If the disease is contagious, cases cluster, independence fails and the variance exceeds the mean.

```
set.seed(42)
n <- 1000
lam <- 3

ind <- rbinom(n, size = 1e6, prob = lam / 1e6)

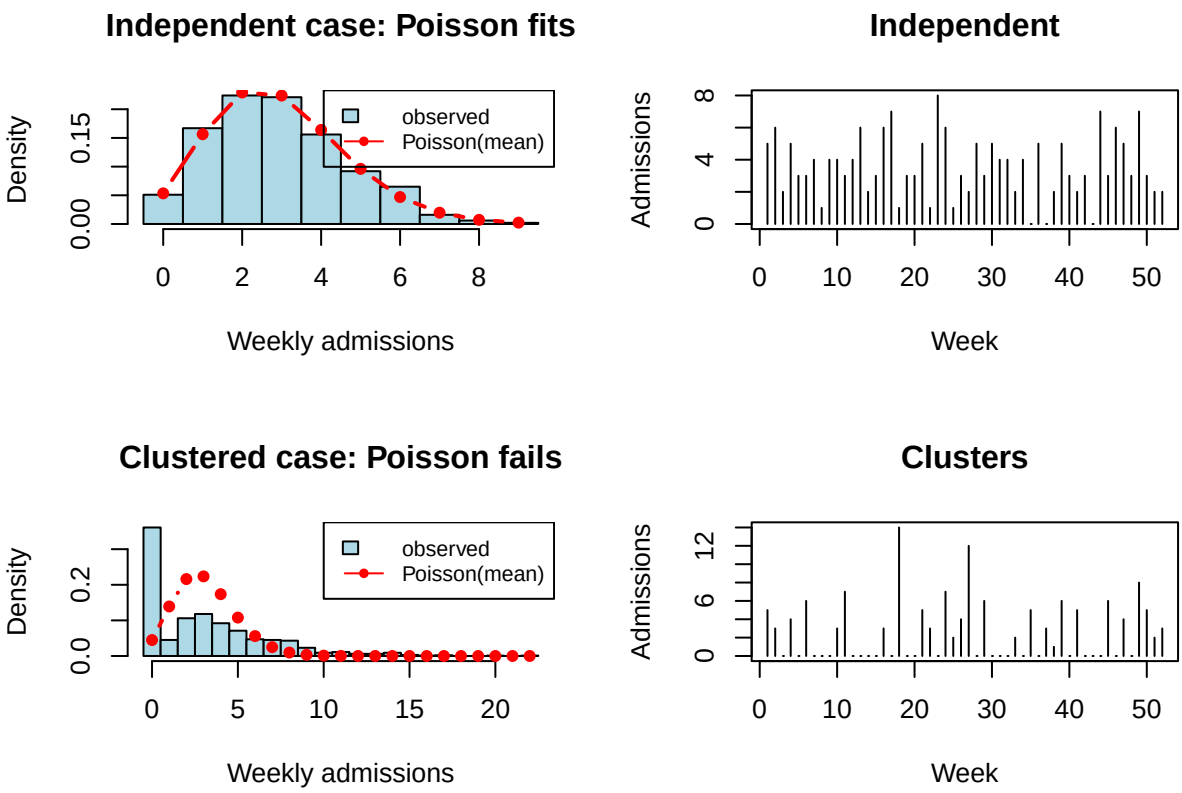
# Clustered cases: Poisson(1) primary cases per week, each bringing Poisson(2) linked cases.
# The mean is still 3, but admissions arrive in clumps.
clu <- replicate(n, {
  k <- rpois(1, 1)
  sum(1 + rpois(k, 2))
})

dis <- function(y) var(y) / mean(y)
round(c(independent = dis(ind), clustered = dis(clu)), 2)
```

```
## independent   clustered
##        1.00        3.84
```

```
show <- function(y, main) {
  m <- max(y)
  hist(y,
    breaks = seq(-0.5, m + 0.5, 1), freq = FALSE, col = "lightblue",
    main = main, xlab = "Weekly admissions"
  )
  points(0:m, dpois(0:m, mean(y)), type = "b", col = "red", pch = 16, lwd = 2)
  legend("topright", c("observed", "Poisson(mean)"),
    fill = c("lightblue", NA), border = c("black", NA),
    col = c(NA, "red"), lty = c(NA, 1), pch = c(NA, 16), cex = 0.8
  )
}

par(mfrow = c(2, 2))
show(ind, "Independent case: Poisson fits")
plot(ind[1:52], type = "h", xlab = "Week", ylab = "Admissions", main = "Independent")
show(clu, "Clustered case: Poisson fails")
plot(clu[1:52], type = "h", xlab = "Week", ylab = "Admissions", main = "Clusters")
```



In the independent case the histogram follows the Poisson curve and var/mean ≈ 1. In the clustered case the mean is the same, but the distribution has more zeros and a heavier right tail, with var/mean ≈ 3.7.

Ex 2.21

Here the text.

Solution

Add comments to the solution.

```
# code here
```

Ex 2.26

Refer to Table 2.4 cross classifying happiness with family income.

(a) Find and interpret the correlation using scores (i) (1, 2, 3) for each variable, (ii) (1, 2, 3) for family income and (1, 4, 5) for happiness.

(b) Construct the joint distribution that has these marginal distributions and exhibits independence of X and Y.

Solution

(a) To calculate the correlation ρ between X (family income) and Y (happiness), we use the expected values and variances of X and Y:

$$\rho = \frac{E[XY] - E[X]E[Y]}{\sqrt{Var(X) \cdot Var(Y)}}$$

(i) Scores $(1, 2, 3)$ for both X and Y

$$\rho \approx 0.1907$$

Weak positive correlation. People with higher family income tend to report slightly higher happiness levels.

(ii) Scores $(1, 2, 3)$ for X and $(1, 4, 5)$ for Y

$$\rho \approx 0.1898$$

The correlation remains almost unchanged.

```
P <- matrix(
  c(
    0.080, 0.198, 0.079,
    0.043, 0.254, 0.143,
    0.017, 0.105, 0.081
  ),
  nrow = 3, byrow = T
)

get_rho <- function(x, y, P) {
  r <- rowSums(P)
  c <- colSums(P)
  Ex <- sum(r * x)
  Ey <- sum(c * y)
  Ex2 <- sum(r * x^2)
  Ey2 <- sum(c * y^2)
  EXY <- sum(P * outer(x, y))

  (EXY - Ex * Ey) / sqrt((Ex2 - Ex^2) * (Ey2 - Ey^2))
}

rho1 <- get_rho(1:3, 1:3, P)
rho2 <- get_rho(1:3, c(1, 4, 5), P)
round(c(rho1, rho2), 4)
```

```
## [1] 0.1907 0.1898
```

(b) Under independence, $P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$.

Multiplying row totals $(0.357, 0.440, 0.203)$ by column totals $(0.140, 0.557, 0.303)$ gives:

$$P_{ind} = \begin{pmatrix} 0.050 & 0.199 & 0.108 \\ 0.062 & 0.245 & 0.133 \\ 0.028 & 0.113 & 0.062 \end{pmatrix}$$

Summing across each row or column restores the original marginal probabilities.

Ex 2.52

Here the text.

Solution

Add comments to the solution.

```r
# code here
```

Ex. 2.53

If Y is a standard normal random variable, with cdf Φ, what is the probability distribution of $X = \Phi(Y)$? Illustrate by randomly generating a million standard normal random variables, applying the cdf function $\Phi()$ to each, and plotting histograms of the (a) y values, (b) x values.

Solution

So $Y \sim \mathcal{N}(0,1)$, with $\Phi(y) = P(Y \leq y)$. Since $\Phi(y)$ returns a probability value, $0 \leq \Phi(y) \leq 1$ and so $0 \leq X \leq 1$. In order to understand how X is distributed in this interval, we can try to compute the distribution function of X, that is to say

$$F_X(x) = P(X \leq x) = P(\Phi(Y) \leq x)$$

We want to isolate Y so we apply the inverse function:

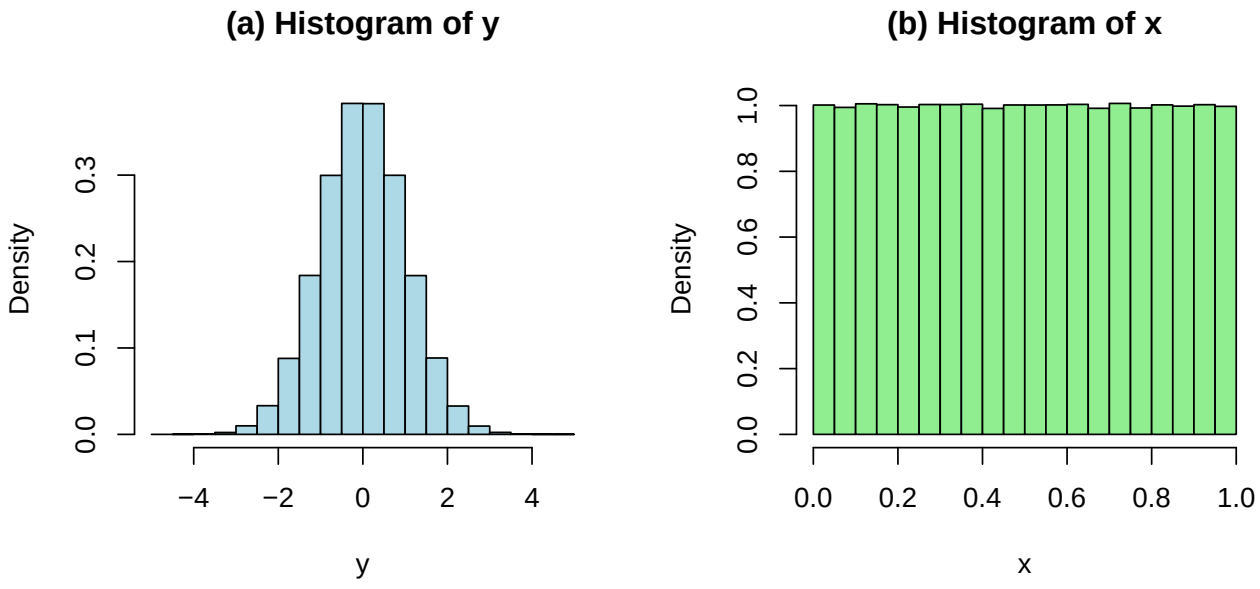
$$F_X(x) = P(X \leq x) = P(Y \leq \Phi^{-1}(x))$$

By definition $P(Y \leq a) = \Phi(a)$ then

$$F_X(x) = \Phi(\Phi^{-1}(x)) = x \sim \mathcal{U}(0,1)$$

```r
set.seed(123)
y <- rnorm(1000000)
x <- pnorm(y)

par(mfrow = c(1, 2))
hist(y, freq = FALSE, main = "(a) Histogram of y", xlab = "y", col = "lightblue")
hist(x, freq = FALSE, main = "(b) Histogram of x", xlab = "x", col = "lightgreen")
```



Comment We can see that histogram (a) clearly shows a Standard Normal distribution with its typical symmetric bell shape centered at 0, while histogram (b) clearly follows a continuous Uniform distribution on $[0,1]$ since all bars have a similar height (density close to 1). This empirically confirms our theoretical result that $X = \Phi(Y) \sim \text{Unif}(0,1)$.

Ex 2.70

The beta distribution is a probability distribution over $(0, 1)$ that is often used in applications for which the random variable is a proportion. The beta pdf is

$$f(y; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1}(1-y)^{\beta-1}, \qquad 0 \le y \le 1$$

for parameters α and β, where $\Gamma(\cdot)$ denotes the gamma function.

(a) Show that the uniform distribution is the special case $\alpha = \beta = 1$.
(b) Show that $\mu = E(Y) = \frac{\alpha}{\alpha+\beta}$.
(c) Find $E(Y^2)$. Show that $\text{var}(Y) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{\mu(1-\mu)}{\alpha+\beta+1}$. For fixed $\alpha + \beta$, note that $\text{var}(Y)$ decreases as μ approaches 0 or 1.
(d) Using a function such as `dbeta` in R, plot the beta pdf for (i) $\alpha = \beta = 0.5, 1.0, 10, 100$, (ii) some values of $\alpha > \beta$ and some values of $\alpha < \beta$. Describe the impact of α and β on the shape and spread.

Solution

(a) The beta pdf with $\alpha = \beta = 1$ is

$$f(y; 1, 1) = \frac{\Gamma(1 + 1)}{\Gamma(1)\Gamma(1)} y^{1-1}(1-y)^{1-1}, \qquad 0 \le y \le 1$$

For a positive integer n, $\Gamma(n) = (n-1)!$, so

$$\begin{aligned} \Gamma(1+1) = \Gamma(2) &= 1! = 1 \\ \Gamma(1) &= 0! = 1 \\ y^{1-1} &= y^0 = 1 \\ (1-y)^{1-1} &= (1-y)^0 = 1 \end{aligned}$$

Thus

$$f(y; 1, 1) = \frac{1!}{0!\,0!} \cdot 1 = 1, \qquad 0 \le y \le 1$$

The uniform pdf on (a, b) with $a = 0$, $b = 1$ is

$$f(y; a, b) = \frac{1}{b-a} = \frac{1}{1-0} = 1, \qquad 0 \le y \le 1$$

```
y <- seq(0, 1, length.out = 5)
dbeta(y, 1, 1)
```

```
## [1] 1 1 1 1 1
```

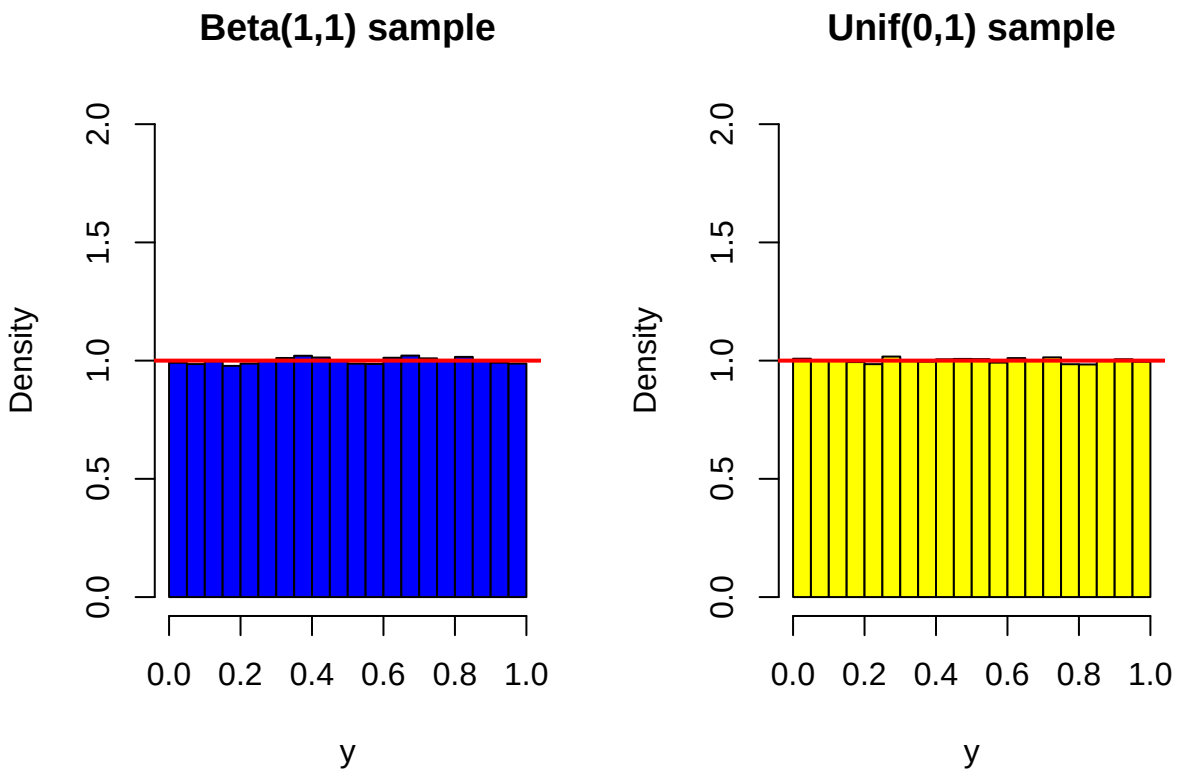
```
dunif(y, 0, 1)
```

```
## [1] 1 1 1 1 1
```

```
par(mfrow = c(1, 2))

n <- 1e5
hist(rbeta(n, 1, 1),
  freq = FALSE, breaks = 30, col = "blue",
  ylim = c(0, 2), xlab = "y", main = "Beta(1,1) sample"
)
abline(h = 1, col = "red", lwd = 2)
```

```r
hist(runif(n, 0, 1),
  freq = FALSE, breaks = 30, col = "yellow",
  ylim = c(0, 2), xlab = "y", main = "Unif(0,1) sample"
)
abline(h = 1, col = "red", lwd = 2)
```



(b) The mean is

$$E[Y] = \int_0^1 y\, f(y;\alpha,\beta)\, dy = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 y^{\alpha}(1-y)^{\beta-1}\, dy$$

Recall the beta and gamma functions

$$B(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}\, dx, \qquad \Gamma(z) = \int_0^\infty t^{z-1}e^{-t}\, dt$$

The product of two gamma functions is

$$\Gamma(a)\Gamma(b) = \int_0^\infty \int_0^\infty u^{a-1}v^{b-1}e^{-(u+v)}\, du\, dv$$

Take $s = u + v \in (0, +\infty)$ and $x = \dfrac{u}{u+v} \in (0,1)$, so that

$$\begin{cases} u = xs \\ v = (1-x)s \end{cases}$$

The change of variables has Jacobian

$$|J| = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial s} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial s} \end{vmatrix} = \begin{vmatrix} s & x \\ -s & 1-x \end{vmatrix} = s(1-x) + sx = s$$

So $du\,dv = s\,dx\,ds$, and the integrand becomes

$$(xs)^{a-1}[(1-x)s]^{b-1}e^{-s}\,s = x^{a-1}(1-x)^{b-1}s^{a+b-1}e^{-s}$$

Thus

$$\Gamma(a)\Gamma(b) = \underbrace{\int_0^1 x^{a-1}(1-x)^{b-1}\,dx}_{B(a,b)} \cdot \underbrace{\int_0^\infty s^{a+b-1}e^{-s}\,ds}_{\Gamma(a+b)} \quad \Rightarrow \quad B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

Finally, with $a = \alpha + 1$ and $b = \beta$,

$$E[Y] = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot B(\alpha+1,\beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}$$

Using $\Gamma(n+1) = n\Gamma(n)$,

$$E[Y] = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\alpha\,\Gamma(\alpha)\Gamma(\beta)}{(\alpha+\beta)\Gamma(\alpha+\beta)} = \boxed{\frac{\alpha}{\alpha+\beta}}$$

(c)

$$E[Y^2] = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 y^{\alpha+1}(1-y)^{\beta-1}\,dy = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\,B(\alpha+2,\beta)$$

This is the beta integral with $a = \alpha + 2$ and $b = \beta$:

$$B(\alpha+2,\beta) = \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)}$$

Using $\Gamma(n+2) = (n+1)n\,\Gamma(n)$,

$$E[Y^2] = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{(\alpha+1)\alpha\,\Gamma(\alpha)\Gamma(\beta)}{(\alpha+\beta+1)(\alpha+\beta)\Gamma(\alpha+\beta)} = \boxed{\frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}}$$

The variance is $V(Y) = E[Y^2] - (E[Y])^2$:

$$\begin{aligned} V(Y) &= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2 \\ &= \frac{\alpha(\alpha+1)(\alpha+\beta) - \alpha^2(\alpha+\beta+1)}{(\alpha+\beta)^2(\alpha+\beta+1)} \\ &= \frac{\alpha^3 + \alpha^2\beta + \alpha^2 + \alpha\beta - \alpha^3 - \alpha^2\beta - \alpha^2}{(\alpha+\beta)^2(\alpha+\beta+1)} \\ &= \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} \end{aligned}$$

With $\mu = \dfrac{\alpha}{\alpha+\beta}$ and $1-\mu = \dfrac{\beta}{\alpha+\beta}$,

$$V(Y) = \frac{\mu\,\beta}{(\alpha+\beta)(\alpha+\beta+1)} = \frac{\mu(1-\mu)}{\alpha+\beta+1}$$

For fixed $\alpha+\beta$, $\mu(1-\mu)$ is largest at $\mu = 1/2$ and tends to 0 as $\mu \to 0$ or 1, so $V(Y)$ decreases as μ approaches 0 or 1.

```r
cols <- c("red", "blue", "green", "yellow", "orange")
x <- seq(1e-6, 0.999999, length.out = 1000)

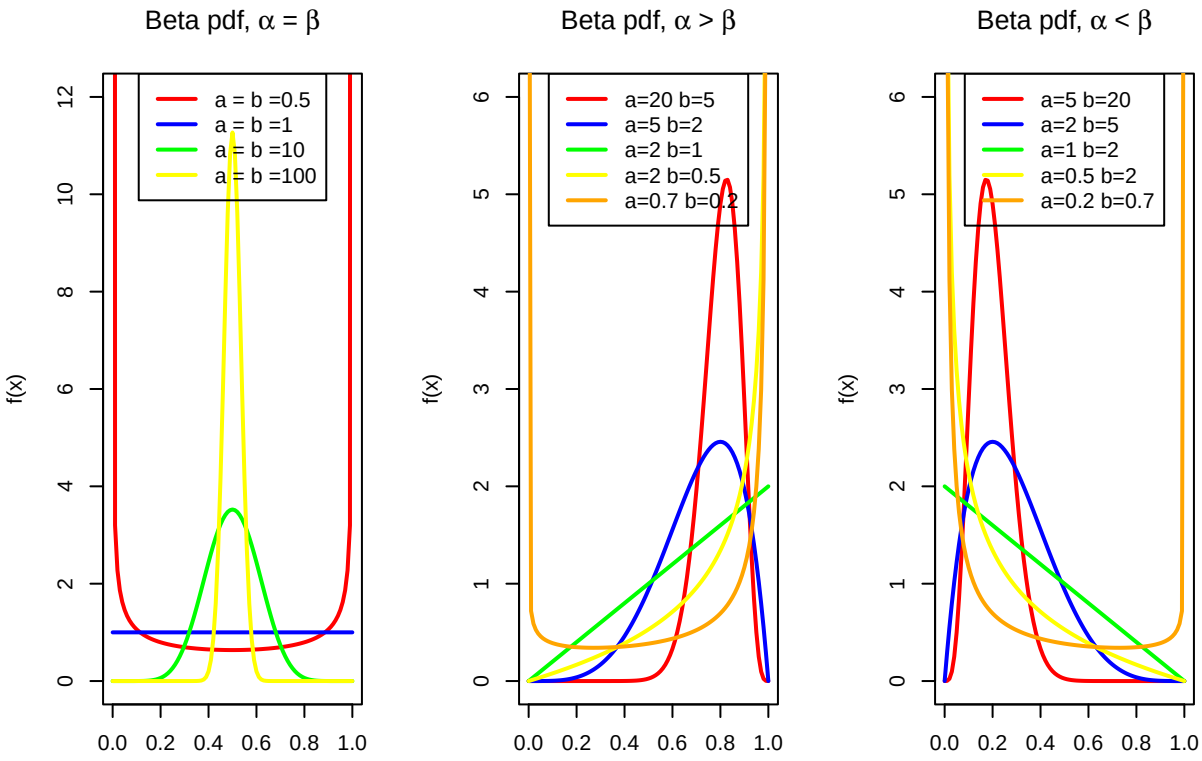
# f <- function(x, a, b){
#   exp(lgamma(a+b) - (lgamma(a) + lgamma(b)) + log(x)*(a-1) + log(1-x)*(b-1))
# }

par(mfrow = c(1, 3))

# (i) alpha = beta
alpha <- c(0.5, 1, 10, 100)
beta <- alpha
curve(dbeta(x, alpha[1], beta[1]), 1e-6, 999999e-6,
  col = cols[1], lwd = 2, ylim = c(0, 12),
  xlab = "x", ylab = "f(x)",
  main = expression(paste("Beta pdf, ", alpha, " = ", beta))
)
for (i in 2:4) curve(dbeta(x, alpha[i], beta[i]), add = TRUE, col = cols[i], lwd = 2)
legend("top", legend = paste0("a = b =", alpha), col = cols, lwd = 2)

# (ii) alpha > beta
ab <- list(c(20, 5), c(5, 2), c(2, 1), c(2, 0.5), c(0.7, 0.2))
curve(dbeta(x, ab[[1]][1], ab[[1]][2]), 1e-6, 999999e-6,
  col = cols[1], lwd = 2, ylim = c(0, 6),
  xlab = "x", ylab = "f(x)",
  main = expression(paste("Beta pdf, ", alpha, " > ", beta))
)
for (i in 2:5) curve(dbeta(x, ab[[i]][1], ab[[i]][2]), add = TRUE, col = cols[i], lwd = 2)
legend("top", legend = sapply(ab, function(p) paste0("a=", p[1], " b=", p[2])), col = cols, lwd = 2)

# (iii) alpha < beta
ab <- list(c(5, 20), c(2, 5), c(1, 2), c(0.5, 2), c(0.2, 0.7))
curve(dbeta(x, ab[[1]][1], ab[[1]][2]), 1e-6, 999999e-6,
  col = cols[1], lwd = 2, ylim = c(0, 6),
  xlab = "x", ylab = "f(x)",
  main = expression(paste("Beta pdf, ", alpha, " < ", beta))
)
for (i in 2:5) curve(dbeta(x, ab[[i]][1], ab[[i]][2]), add = TRUE, col = cols[i], lwd = 2)
legend("top", legend = sapply(ab, function(p) paste0("a=", p[1], " b=", p[2])), col = cols, lwd = 2)
```



(d)

Shape

- α controls the left edge, β the right edge. Near 0, $f \approx x^{\alpha-1}$; near 1, $f \approx (1-x)^{\beta-1}$. If the parameter is below 1, the density blows up at that edge. At 1 it stays finite and nonzero. Above 1 it goes to 0.
- **Both below 1:** U-shaped, with mass piled at both ends.
- **Both equal to 1:** uniform.
- **Both above 1:** a single hump, with mode $\dfrac{\alpha-1}{\alpha+\beta-2}$.
- $\alpha = \beta$: symmetric about 0.5.
- $\alpha > \beta$: mass shifts toward 1.
- $\alpha < \beta$: mass shifts toward 0.

The ratio sets the mean:

$$E[Y] = \frac{\alpha}{\alpha+\beta}$$

Doubling both α and β leaves the mean unchanged. Only the balance between them moves the centre.

Spread The total $\alpha+\beta$ sets the concentration:

$$\mathrm{Var}(Y) = \frac{\mu(1-\mu)}{\alpha+\beta+1}, \qquad \mu = \frac{\alpha}{\alpha+\beta}$$

For a fixed mean, a larger $\alpha+\beta$ gives a smaller variance. Small totals (0.5 + 0.5) give a spread-out, U-shaped density, and large totals (100 + 100) give a narrow, near-normal peak.

FSDS - Chapter 3

Ex 3.18

Here the text.

Solution

Add comments to the solution.

```
# code here
```

Ex 3.28

A survey is planned to estimate the population proportion π supporting more government action to address global warming. For a simple random sample, if π may be near 0.50, how large should n be so that the standard error of the sample proportion ($\hat{\pi}$) is 0.04?

Solution

The standard error formula for a sample proportion is:

$$SE(\hat{\pi}) = \sqrt{\frac{\pi(1-\pi)}{n}}$$

- Target Standard Error: $SE(\hat{\pi}) = 0.04$
- Estimated proportion: $\pi = 0.50$

$$\sqrt{\frac{0.50(1-0.5)}{n}} = \sqrt{\frac{1}{4n}} = 0.04$$

$$n = \frac{1}{4 \cdot 0.04^2} = 156.25$$

Since sample size must be a whole number of individuals:

$$n = \lceil 156.25 \rceil = 157$$

Ex 3.24

Here the text.

Solution

Add comments to the solution.

```
# code here
```

FSDS - Chapter 4

Ex 4.14

Using the Students data file, for the corresponding population, construct a 95% confidence interval

(a) for the mean weekly number of hours spent watching TV;

(b) to compare females and males on the mean weekly number of hours spent watching TV.

In each case, state assumptions, including the practical importance of each, and interpret results.

Solution

We start loading the data.

```
data <- read.table("http://stat4ds.rwth-aachen.de/data/Students.dat", header = TRUE)
head(data)
```

```
##   subject gender age hsgpa cogpa dhome dres tv sport news aids veg affil ideol
## 1       1      0  32   2.2   3.5     0  5.0  3     5    0    0   0     2     6
## 2       2      1  23   2.1   3.5  1200  0.3 15     7    5    6   1     1     2
## 3       3      1  27   3.3   3.0  1300  1.5  0     4    3    0   1     1     2
## 4       4      1  35   3.5   3.2  1500  8.0  5     5    6    3   0     3     4
## 5       5      0  23   3.1   3.5  1600 10.0  6     6    3    0   0     3     1
## 6       6      0  39   3.5   3.5   350  3.0  4     5    7    0   1     1     2
##   relig abor affirm life
## 1     2    0      0    1
## 2     1    1      1    3
## 3     2    1      1    3
## 4     1    1      1    2
## 5     0    1      0    2
## 6     1    1      1    3
```

(a) In order to make a 95% CI for the mean of the population μ when σ is not known, we use the Student's t:

$$\bar{y} \pm t_{0.025,df} \cdot \frac{s}{\sqrt{n}}$$

```
CI_95 <- t.test(data$tv, conf.level = 0.95)
CI_95
```

```
##
##  One Sample t-test
##
## data:  data$tv
## t = 8.3794, df = 59, p-value = 1.252e-11
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  5.531395 9.001938
## sample estimates:
## mean of x
##  7.266667
```

So the sample mean is $\bar{Y} = 7.266667$ and the CI is $[5.531395, 9.001938]$.

The fundamental assumptions for this interval are:

1. The 60 students must be a representative random sample of the target population. Id the sample is distorted, the inference cannot be generalised.
2. The TV hour distribution should be Normal. The one-sample t-test is very robust to deviations from normality when the sample size is not small (here $n = 60 \geq 30$), thanks to the CLT.

(b) compare females and males We make now a 95% CI for the difference between the mean of the 2 grups ($\mu_{females} - \mu_{males}$) using `TV` and `gender` columns.

```
CI_d95 <- t.test(data$tv ~ data$gender)
CI_d95
```

```
##
##  Welch Two Sample t-test
##
## data:  data$tv by data$gender
## t = -0.84995, df = 56.249, p-value = 0.399
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
```

```
## 95 percent confidence interval:
##  -4.980854  2.013112
## sample estimates:
## mean in group 0 mean in group 1
##        6.500000        7.983871
```

The fundamental assumptions for this interval are:

1. The two groups (males and females) must be drawn from independent random samples. Practical significance: this is essential for the validity of the inference.
2. The variable `tv` should be normally distributed in both groups. Here too, the two-sample t-test is robust to non-normality thanks to the size of the two subsamples (around 30 per group). (Note: by using Welch's test, we do not need to assume that the variances are equal, i.e. $\sigma_0^2 = \sigma_1^2$, which protects us from further errors).

- **Interpretation:** The 95% confidence interval for the difference in population means ($\mu_0 - \mu_1$, males minus females) is (-4.98, 2.01) hours. Since this interval contains 0 (and the p-value is $0.399 > 0.05$), we fail to reject $H_0 : \mu_0 = \mu_1$. We conclude that there is no statistically significant difference between males and females in the mean weekly hours spent watching TV (males could watch up to ~5 hours less or up to ~2 hours more than females on average)

Ex 4.16

The Substance data file at the book's website shows a contingency table formed from a survey that asked a sample of high school students whether they have ever used alcohol, cigarettes, and marijuana. Construct a 95% Wald confidence interval to compare those who have used or not used alcohol on whether they have used marijuana, using (a) formula (4.13); (b) software. State assumptions for your analysis, and interpret results.

Solution

Collapsing over cigarettes, the table is

Alcohol	Marijuana yes	Marijuana no	Total
yes	955	994	1949
no	5	322	327

Assumptions

- The students who have used alcohol and those who have not form two **independent binomial samples**, and each student's response is independent of the others.
- The sample behaves like a random sample from the population of interest.
- The counts are large enough for the normal approximation. All cells are at least 5, but the 5 marijuana users among non-drinkers is borderline.
- Cigarette use is ignored by collapsing, which could confound the comparison.

(a) By hand The Wald interval for $p_1 - p_2$ is

$$(\hat{p}_1 - \hat{p}_2) \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

with

$$\hat{p}_1 = \frac{955}{1949} = 0.4900, \qquad \hat{p}_2 = \frac{5}{327} = 0.0153, \qquad z_{0.975} = 1.96$$

$$SE = \sqrt{\frac{0.4900(0.5100)}{1949} + \frac{0.0153(0.9847)}{327}} = 0.0132$$

$$CI = 0.4747 \pm 1.96(0.0132) = 0.4747 \pm 0.0259 \Rightarrow (0.449,\ 0.501)$$

```
substance <- read.table("http://stat4ds.rwth-aachen.de/data/Substance.dat", header = TRUE)

tab <- tapply(substance$count, list(substance$alcohol, substance$marijuana), sum)
tab
```

(b) Software

```
##      no yes
## no  322   5
## yes 994 955
```

```
success <- tab[c("yes", "no"), "yes"]
total <- rowSums(tab)[c("yes", "no")]

p_hat <- success / total
d <- p_hat[1] - p_hat[2]
se <- sqrt(sum(p_hat * (1 - p_hat) / total))
d + c(-1, 1) * qnorm(0.975) * se
```

```
## [1] 0.4488310 0.5005777
```

```
prop.test(success, total, correct = FALSE)$conf.int
```

```
## [1] 0.4488310 0.5005777
## attr(,"conf.level")
## [1] 0.95
```

Interpretation We are 95% confident that the proportion who have used marijuana is between 0.449 and 0.501 higher among students who have used alcohol than among those who have not.

Ex 4.48

Here the text.

Solution

Add comments to the solution.

```
# code here
```

FSDS - Chapter 5

Ex 5.2

When a government does not have enough money to pay for the services that it provides, it can raise taxes or it can reduce services. When the Florida Poll asked a random sample of 1200 Floridians which they preferred, 52% (624 of the 1200) chose *raise taxes* and 48% chose reduce services.

Let π denote the population proportion of Floridians who would choose raising taxes. Analyze whether this is a minority of the population ($\pi < 0.50$) or a majority ($\pi > 0.50$) by testing $H_0 : \pi = 0.50$ against

$H_a : \pi \neq 0.50$. Interpret the P-value. Is it appropriate to "accept H_0? Why or why not?

Solution

$n = 1200 \quad X = 624 \quad \hat{\pi} = 0.52$

Test:

$$H_0 : \pi = 0.50$$
$$H_a : \pi \neq 0.50$$

Under the null hypothesis, the standard error is calculated as:

$$\text{SE}_0(\hat{\pi}) = \sqrt{\frac{0.50 \times 0.50}{1200}} \approx 0.01443$$

This gives a Wald test statistic of:

$$Z = \frac{0.52 - 0.50}{0.01443} \approx 1.386$$

The corresponding two-sided p-value is $2 \times P(Z \geq 1.386) \approx 0.166$. Because $p = 0.166 > 0.05$, there is insufficient statistical evidence to reject H_0. The observed 52% preference can be naturally accounted for by random sampling variation.

We avoid stating that we "accept" H_0 because a non-significant result merely shows that the observed data are compatible with the null hypothesis, not that $\pi = 0.50$ is proven true. Frequentist hypothesis tests control for Type I errors rather than proving null parameters, so a high p-value might simply reflect limited power to detect small deviations.

Ex 5.12

Here the text.

Solution

Add comments to the solution.

```
# code here
```

Ex 5.50

A random sample of size 40 has $\bar{y} = 120$. The P-value for testing $H_0 : \mu = 100$ against $H_a : \mu \neq 100$ is 0.057. Explain what is incorrect about each of the following interpretations of this P-value, and provide a proper interpretation.

(a) The probability that H_0 is correct equals 0.057.

(b) The probability that $\bar{y} = 120$ if H_0 is true equals 0.057.

(c) The probability of Type I error equals 0.057.

(d) We can accept H_0 at the $\alpha = 0.05$ level.

Solution

- **(a) Incorrect:** In classical (frequentist) inference, the population parameter μ is fixed, not a random variable, so H_0 is either true (probability 1) or false (probability 0). The P-value is the probability of observing extreme data given that H_0 is true, $P(\text{extreme data} \mid H_0)$, not the probability that H_0 is true given the data, $P(H_0 \mid \text{data})$.
- **(b) Incorrect:** For a continuous sampling distribution, the probability of observing the exact value $\bar{y} = 120$ is 0. The P-value refers to the probability of observing a sample mean **at least as extreme as** 120 in either tail (i.e., $\bar{Y} \geq 120$ or $\bar{Y} \leq 80$).
- **(c) Incorrect:** The probability of a Type I error is the preset significance level α (e.g., $\alpha = 0.05$), not the P-value, which varies from sample to sample. Moreover, if we do not reject H_0 (since $0.057 > 0.05$), we cannot commit a Type I error at all—we only risk committing a **Type II error**.
- **(d) Incorrect:** When the P-value is greater than $\alpha = 0.05$, we **fail to reject** H_0 rather than "accepting" it. Failing to find sufficient evidence against $\mu = 100$ does not prove that μ is equal to 100; many other values around 100 are also plausible.
- **Proper Interpretation:** Assuming that the null hypothesis $H_0 : \mu = 100$ is true, the probability of obtaining a sample mean $\bar{Y}$ at least 20 units away from 100 in either direction (that is, $\bar{Y} \geq 120$ or $\bar{Y} \leq 80$) in a random sample of size 40 is 0.057.

CS - Chapter 1

Ex 1.1

An exponential random variable $X \geq 0$ has p.d.f. $f(x) = \lambda e^{-\lambda x}$, with $\lambda > 0$.

1. Find the c.d.f. and the quantile function of X.
2. Find $P(X \leq \lambda)$ and the median of X.
3. Find the mean and variance of X.

Solution

(1) The c.d.f. is obtained by integrating up to x. Since $\dfrac{d}{dt}\left(-e^{-\lambda t}\right) = \lambda e^{-\lambda t}$,

$$F(x) = \int_0^x \lambda e^{-\lambda t}\, dt = \Big[-e^{-\lambda t} \Big]_0^x = \boxed{1 - e^{-\lambda x}}, \qquad x \geq 0$$

A check that f is a valid density:

$$\int_0^\infty \lambda e^{-\lambda x}\, dx = \Big[-e^{-\lambda x} \Big]_0^\infty = 0 + 1 = 1$$

The quantile function solves $F(x) = p$ for x, with $p \in (0, 1)$:

$$\begin{aligned} 1 - e^{-\lambda x} &= p \\ e^{-\lambda x} &= 1 - p \\ -\lambda x &= \ln(1-p) \\ x &= -\frac{\ln(1-p)}{\lambda} \end{aligned} \qquad \Rightarrow \qquad \boxed{Q(p) = -\frac{\ln(1-p)}{\lambda}}$$

(2) Substituting $x = \lambda$ in the c.d.f.,

$$P(X \leq \lambda) = F(\lambda) = 1 - e^{-\lambda \cdot \lambda} = \boxed{1 - e^{-\lambda^2}}$$

The median is the 0.5 quantile:

$$Q(0.5) = -\frac{\ln(0.5)}{\lambda} = \boxed{\frac{\ln 2}{\lambda}}$$

(3) Mean. $E[X] = \int_0^\infty x\,\lambda e^{-\lambda x}\,dx$. Integrate by parts with $u = x$ and $dv = \lambda e^{-\lambda x}dx$, so $du = dx$ and $v = -e^{-\lambda x}$:

$$\begin{aligned} E[X] &= \Big[-xe^{-\lambda x}\Big]_0^\infty + \int_0^\infty e^{-\lambda x}\,dx \\ &= 0 + \frac{1}{\lambda}\int_0^\infty \lambda e^{-\lambda x}\,dx \\ &= \boxed{\frac{1}{\lambda}} \end{aligned}$$

The boundary term vanishes because $xe^{-\lambda x} \to 0$ as $x \to \infty$ (exponential decay beats linear growth).

Second moment. $E[X^2] = \int_0^\infty x^2\,\lambda e^{-\lambda x}\,dx$. Integrate by parts with $u = x^2$ and $dv = \lambda e^{-\lambda x}dx$:

$$\begin{aligned} E[X^2] &= \Big[-x^2e^{-\lambda x}\Big]_0^\infty + \int_0^\infty 2x\,e^{-\lambda x}\,dx \\ &= 0 + \frac{2}{\lambda}\int_0^\infty x\,\lambda e^{-\lambda x}\,dx \\ &= \frac{2}{\lambda}\,E[X] = \boxed{\frac{2}{\lambda^2}} \end{aligned}$$

Variance.

$$V(X) = E[X^2] - (E[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \boxed{\frac{1}{\lambda^2}}$$

Ex 1.6

Here the text.

Solution

Add comments to the solution.

```
# code here
```

CS - Chapter 3

Ex 3.5

Consider solving the matrix equation $Ax = y$ for x, where y is a known n-vector and A is a known $n \times n$ matrix. The formal solution to the problem is $x = A^{-1}y$, but it is possible to solve the equation directly, without actually forming A^{-1}.

a. First create an A, x and y satisfying $Ax = y$.
b. Using `solve`, form the matrix A^{-1} explicitly and then form $x_1 = A^{-1}y$. Note how long this takes and assess the mean absolute error.
c. Now use `solve` to directly solve for x without forming A^{-1}. Note how long this takes and assess the mean absolute error.
d. What do you conclude?

Solution

Add comments to the solution.

```
set.seed(0)
n <- 1000
A <- matrix(runif(n * n), n, n)
```

```r
x.true <- runif(n)
y <- A %*% x.true

t1 <- system.time({
  A_inv <- solve(A)
  x1 <- A_inv %*% y
})
mae1 <- mean(abs(x1 - x.true))

t2 <- system.time({
  x2 <- solve(A, y)
})
mae2 <- mean(abs(x2 - x.true))

cat("Explicit inverse : Time =", t1["elapsed"], "s , MAE =", mae1, "\n")
```

```
## Explicit inverse : Time = 0.555 s , MAE = 2.956833e-11
```

```r
cat("Direct solution  : Time =", t2["elapsed"], "s , MAE =", mae2, "\n")
```

```
## Direct solution  : Time = 0.167 s , MAE = 1.356142e-12
```

Directly solving the linear system using `solve(A, y)` is strongly preferred: it is significantly faster and numerically more accurate (having a lower mean absolute error) than explicitly inverting the matrix via `solve(A) %*% y`.

Ex 3.6

Here the text.

Solution

Add comments to the solution.

```r
# code here
```